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Molecular and Functional Analysis of Calcium Binding by a Cancer-linked Calreticulin Mutant

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eLife Assessment

This study investigates low-affinity Ca²⁺ binding by WT calreticulin and mutant calreticulin associated with type I myeloproliferative neoplasms, as well as the impact on Ca²⁺ fluxes in suspension cultures of megakaryocyte-like cells in vitro in response to ER Ca²⁺ ATPase inhibitors that deplete endoplasmic reticulum (ER) Ca²⁺ store and open plasma membrane Ca²⁺ channels through STIM1-Orai interactions. The results are **important** in that they show that Ca²⁺ binding by calreticulin and store-operated Ca²⁺ entry are not fundamentally impacted by the type I deletion mutation in calreticulin, which rules out a direct effect of the calreticulin mutation on its own low-affinity Ca²⁺ binding and any broad impact on ER Ca²⁺ regulation. The strength of the data and methods used ranges from **solid** to **convincing**, although the use of suspension-based flow cytometric assays to investigate ER Ca²⁺ levels and Ca²⁺ entry can be challenged. High-affinity Ca²⁺ binding sites could be further considered, and possible confounding effects of Abl kinase activity in the megakaryocyte-like cell lines could be offset.

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Abstract

Calreticulin (CRT) is an endoplasmic reticulum (ER) chaperone with low affinity calcium binding sites in its C-terminal domain. This region is altered by somatic mutations in the CRT gene (*CALR*), which drive a subset of myeloproliferative neoplasms (MPN). Perturbations in ER calcium storage and signaling are reported for the MPN type I mutant, CRT_{Del52}, and are linked to disease pathogenesis. Using recombinant CRT proteins, we found similar low affinity calcium binding characteristics for wild-type CRT and CRT_{Del52}, as determined by isothermal titration calorimetry (ITC). Residues 340–349, conserved in both wild-type CRT and CRT_{Del52}, contribute to binding. Furthermore, ER and cytosolic calcium levels and store-operated calcium entry (SOCE) were comparable in CRT knockout (CRT-KO) cells reconstituted with wild-type CRT or CRT_{Del52}. Notably, the CRT-KO induces expression of multiple ER proteins known to contain low affinity calcium binding sites or regulate ER calcium levels. Overall, these findings indicate that CRT_{Del52} retains at least some low affinity calcium binding capacity and that CRT-deficiency induces compensatory cellular changes that maintain ER calcium homeostasis.

Summary

Pathogenic CRT_{Del52} retains at least partial low-affinity calcium-binding capacity. Additionally, live-cell imaging and flow cytometry with ratiometric calcium probes indicate that ER and cytosolic calcium levels, as well as store-operated calcium entry (SOCE), are comparable in CRT-

deficient cells rescued with either wild-type CRT or CRT_{Del52}.

Introduction

Calreticulin (CRT) is an endoplasmic reticulum (ER) luminal protein that plays key roles in the assembly and folding of N-linked glycoproteins. CRT has a highly acidic C-terminal domain that contains multiple low affinity calcium binding sites (Baksh and Michalak, 1991 [↗](#), Wijeyesakere et al., 2011 [↗](#)). As an ER chaperone, CRT is thought to play a critical role in maintaining cellular calcium homeostasis (Venkatesan et al., 2021b [↗](#), Michalak, 2024 [↗](#)). The ER has a relatively high free calcium concentration of 0.5-1 μ M through the presence of various low affinity calcium-binding proteins that buffer calcium, whereas cytosolic calcium levels are maintained within a narrow range of around 100 nM (Michalak, 2024 [↗](#)). The maintenance of calcium homeostasis within cellular compartments is essential for cell function and survival (Bagur and Hajnoczky, 2017 [↗](#)). Early studies have shown that the C-domain of CRT contributes to ER calcium storage, although free ER luminal concentrations are unchanged (Nakamura et al., 2001 [↗](#)).

Somatic mutations of exon 9 of the *CALR* gene including a 52-base pair deletion (Del52; type 1 mutation) and a 5-base pair insertion (Ins5; type 2 mutation), are driver mutations in some myeloproliferative neoplasms (MPN), specifically in essential thrombocythemia (ET) and primary and secondary myelofibrosis (PMF) (Klampfl et al., 2013 [↗](#), Nangalia et al., 2013 [↗](#)). The MPN mutations change the CRT C-domain residues from being predominantly acidic to a high prevalence of basic amino acids and result in a loss of the ER-retention KDEL sequence (Klampfl et al., 2013 [↗](#), Nangalia et al., 2013 [↗](#)). Although all MPN CRT mutants change the sequence and charge of the C-domain, a greater number of acidic residues are retained in MPN type 2 mutant (CRT_{Ins5}), compared to the type I mutant (CRT_{Del52}). The loss of more acidic residues from the C-terminal domain of CRT_{Del52} has been reported to impair its calcium binding and reduce ER calcium levels (Ibarra et al., 2022 [↗](#)). Store-operated calcium entry (SOCE) is a physiological process that allows for the entry of extracellular calcium and ER calcium restoration following ER calcium depletion (Putney, 2011 [↗](#), Zhang et al., 2020 [↗](#)). Altered calcium release from the ER and SOCE was reported for MPN type I but not type 2 mutants in cultured megakaryocytes from MPN patients (Pietra et al., 2016 [↗](#)). In another study, spontaneous calcium spikes and increased SOCE activation were observed in cultured megakaryocytes from MPN type I and type 2 patients when compared to healthy donors (Di Buduo et al., 2020 [↗](#)).

Although global calcium binding defects were predicted or qualitatively measured for CRT_{Del52} (Shivarov et al., 2014 [↗](#), Ibarra et al., 2022 [↗](#)) and altered cellular calcium signaling is also reported (Pietra et al., 2016 [↗](#), Di Buduo et al., 2020 [↗](#)), it remains unclear whether MPN-linked CRT_{Del52} has any low affinity calcium binding capacity, given that a few conserved acidic residues are retained in its C-domain (Figure 1 [↗](#)). In fact, our previous studies showed that a truncated version of murine wild-type CRT (mCRT) containing the non-mutated part of CRT_{Del52} and two additional C-terminal residues (mCRT₁₋₃₅₁) showed similar low affinity calcium binding characteristics as the wild-type mCRT (Wijeyesakere et al., 2011 [↗](#)). On the other hand, further truncation of mCRT (mCRT₁₋₃₃₉) was needed to disrupt calcium binding (Wijeyesakere et al., 2011 [↗](#), Wijeyesakere et al., 2016 [↗](#)). The goal of the current study was to quantitatively measure low affinity calcium binding by CRT_{Del52} compared with wild-type human CRT and also quantify ER and cytosolic calcium levels and calcium responsiveness in cells expressing wild-type CRT or CRT_{Del52}.

Material and methods

Protein expression and purification

B1 domain of streptococcal protein G (GB1) and his-tagged versions of wild-type CRT, CRT₍₁₋₃₅₁₎, CRT₍₁₋₃₃₉₎ and CRT_{Del52} were purified following expression in *E. coli* cells using a nickel resin and gel filtration chromatography as described previously (DelProposto et al., 2009 [↗](#), Del Cid et al., 2010 [↗](#), Wijeyesakere et al., 2011 [↗](#)) (also see supplemental methods). The monomer fractions

Figure 1. Sequence alignments of the C-terminal residues of murine and human calreticulin (mCRT and hCRT), MPN-linked CRT mutants Ins5 and Del52.

Multiple sequence alignment using Clustal Omega was employed to compare C-domain sequences of mature mCRT, hCRT, Del52, and Ins5 proteins. Acidic residues (340 – 349) previously implicated in low affinity calcium binding by mCRT that are shared between wild-type and both CRT_{Del52} and CRT_{Ins5} are highlighted in green. Additional acidic residues shared just between wild-type CRT and CRT_{Ins5} are shown in red. There are additional acidic residues unique to wild-type CRT. Mature protein numbering is used for the CRT sequences.

were pooled, dialyzed and concentrated to 4 mg/mL in buffer containing 20 mM HEPES, 10 mM NaCl 1 mM CaCl_2 pH 7.5 using Amicon ultra concentrators with a molecular weight cut-off of 30,000 Dalton (Millipore) and frozen in 10% glycerol for further use.

Isothermal titration calorimetry

Isothermal titration calorimetry (ITC) measurements were done using the Nano-ITC SV (TA Instruments). Purified CRT monomers were dialyzed into 20 mM HEPES, 10 mM NaCl pH 7.5, brought to a concentration of 100 μM , and then pre-incubated with 50-100 μM CaCl_2 in 20 mM HEPES, 10 mM NaCl pH 7.5 for 30 minutes to block the high affinity calcium binding sites. This was followed by 25 sequential 10 μL injections of 20 mM HEPES, 10 mM NaCl, 10 mM CaCl_2 pH 7.5 into the protein solutions maintained at 37°C. The heat changes measured for the sequential injections were curve-fitted to calculate the enthalpies, stoichiometries, and the dissociation constants using NanoAnalyze software.

Cytosolic calcium measurement and store-operated calcium entry (SOCE) induction

MEG-01 cells and their *CALR*-modified versions (see supplemental methods) were plated on glass coverslips pretreated with 0.01% poly-L-lysine to facilitate cell attachment. Cells were loaded with 2.5 μM Fura-2/AM dye (Ion Biosciences) supplemented with 0.1% pluronic acid in a physiological salt solution (PSS) containing 140 mM NaCl, 2 mM CaCl_2 , 5 mM KCl, 2 mM MgCl_2 , 10 mM HEPES and 11 mM glucose, and incubated for 60 minutes at 37°C. Live single-cell imaging was done at 20X magnification using an Olympus IX71 inverted microscope equipped with a TILL Polychrome V monochromator light source, a Dual-View emission beam splitter, and a cooled Photometrics Quant-EM camera. Cells were transferred to a perfusion chamber that was mounted on the microscope stage, and PSS with or without 2 mM CaCl_2 was flowed over cells using a gravity-fed perfusion system. Cytosolic calcium measurements were performed by perfusing Fura-2 loaded cells in the chamber with 2 mM Ca^{2+} , followed by the addition of 25 μM cyclopiazonic acid (CPA) (Cayman) into the imaging buffer. To determine cytosolic calcium levels after activating SOCE, cells were initially perfused with Ca^{2+} -free PSS containing 0.1 mM EGTA, followed by the addition of 25 μM CPA in Ca^{2+} -free PSS and subsequent addition of 2 mM Ca^{2+} and 25 μM CPA in PSS (without EGTA) to reestablish calcium influx. The emission signals corresponding to the 340 and 380 nm excitation were recorded, and the signals resulting from 340/380 nm excitation were calculated using the Meta Fluor software. The basal cytosolic calcium was defined as the 340/380 nm signal ratio before CPA treatment (measured over a 4-or 5-minute interval immediately before CPA addition). The fractional increase in cytosolic calcium levels after ER Ca^{2+} depletion with CPA was calculated as the difference between peak signal (single time point) and baseline (averaged over a 4-or 5-minute interval before CPA addition), as a ratio relative to the baseline. The fractional increase in cytosolic calcium after addition of 2 mM extracellular Ca^{2+} and CPA was calculated as the difference between the peak signal following Ca^{2+} and CPA addition (averaged over a 1-minute time interval following the peak signal) and the baseline (the signal averaged over a 0.8-1-minute interval immediately before the Ca^{2+} and CPA addition) relative to the baseline. Total calcium release after ER Ca^{2+} depletion was reported as the total area under the curve (AUC). Fractional increases were calculated as (Peak-Baseline)/Baseline x 100. This was calculated as peak signals after CPA addition relative to the baseline before CPA (measured over a 4-or 5-minute interval immediately before CPA addition).

ER calcium measurements in HEK293T cells

CALR-modified HEK cells were prepared as described in supplemental methods. Cells were transiently transfected with pCIS-GEM-CEPIA1er (Suzuki et al., 2014 [DOI](#)); Addgene plasmid # 58217) using Fugene HD reagent (Promega). The transfected cells were harvested after 72 hours. Cells were washed twice with FACS buffer (PBS + 2% FBS) supplemented with 2 mM Ca^{2+} and resuspended in the same FACS buffer. The live-dead stain 7-AAD (BD) was added at 1:200 dilution and the cells were analyzed by flow cytometry using the BD LSRFortessa cell analyzer. The signal

for the Ca^{2+} -bound GEM-CEPIA1er probe was recorded in the Pacific Blue channel (excitation at 405 nm and emission at 452-455 nm) and the signal for the unbound probe was recorded in the AmCyan channel (excitation at 405 and emission at 498 nm). The analysis of flow cytometry data was done in the FlowJo software (v10.10.00). The cells were gated based on the forward (FSC-A) and side scatter (SSC-A), followed by gating on the 7AAD-negative (live) cells, and subsequent gating on the cells positive for the expression of GEM-CEPIA1er probe measured in the Pacific Blue and AmCyan channels. Basal signals were initially recorded, followed by the addition of 5 μM thapsigargin (Tg) and subsequent recording of signals for the treated and untreated cells at four different time points between 2 – 10 minutes. The ratio of mean fluorescence intensity (MFI) of the bound to the unbound signal was used for calculations.

RNA-sequencing

The parental and CRT-KO MEG-01 cells were grown in complete media for 12-18 hours before harvesting. Total RNA was isolated using the RNeasy Plus Mini Kit (Qiagen). This was followed by poly-A selection and enrichment of mRNA using NEB Next Ultra Express RNA kit and a NEB Poly A kit (New England Biolabs). Sequencing was done on the NovaSeq at PE150, targeting 30-40 million reads/sample after cDNA library preparation. Data were pre-filtered to remove genes with 0 counts in all samples. Differential gene expression analysis was performed using DESeq2. The differentially expressed genes were selected as those with a log2 fold-change of <-0.49 or >0.49 and p values <0.05 . The results from RNA sequencing were further analyzed in the iPathwayGuide to identify significantly altered proteins related to calcium signaling. The volcano plot was plotted in GraphPad Prism as Log2 fold-change (x-axis) vs. $-\text{Log}_{10}$ adjusted p-value.

Real-time polymerase chain reaction

Total RNA was isolated using the RNeasy Plus Mini Kit (Qiagen) followed by synthesis of complementary DNA using the Superscript III First-Strand Synthesis Kit (ThermoFisher Scientific) following the manufacturer's protocol. Quantitative RT-PCR was performed on the ABI 7500 FAST Real Time PCR machine (ThermoFisher Scientific, USA) using the SYBR Green PCR Master mix (Applied Biosystems, UK) and gene-specific primers (Supplementary Table 2 [↗](#)). Measurement was done in technical duplicates. The expression of target genes was quantified using the $2^{-\Delta\Delta\text{Ct}}$ method and normalized to the expression of either one or multiple endogenous control genes including beta-actin (ACTB), glyceraldehyde 3-phosphate dehydrogenase (GAPDH) and hypoxanthine phosphoribosyltransferase 1 (HPRT1). The expression of the target gene in MEG-01 CRT-KO and wild type and CRT_{Del52} reconstitution cells was further normalized to the mean expression of parental MEG-01 or CRT-KO-vec. The calculated relative mRNA transcript expression values (RQ) were plotted using GraphPad Prism.

Results

MPN-linked CRT_{Del52} retained at least some low affinity calcium binding sites

Previous studies (Wijeyesakere et al., 2011 [↗](#)) have implicated residues 340 to 351 of murine CRT, which are identical in wild-type human CRT and fully conserved with the 340 to 349 residues of CRT_{Del52}, to be involved in low affinity calcium binding. Further truncation of the C-domain to the residue 339 (mCRT₁₋₃₃₉) led to the loss of calcium binding measured by ITC (Wijeyesakere et al., 2011 [↗](#)). Figure 1 [↗](#) shows a sequence alignment of murine CRT, human CRT, CRT_{Ins5}, and CRT_{Del52} proteins and indicates that the residues previously implicated in low affinity calcium binding are also shared between wild-type CRT and CRT_{Del52}. Recombinant his and GB1 tagged purified proteins were used to compare calcium binding by human CRT and its variants. Chromatograms for human wild-type CRT, CRT_{Del52}, CRT₍₁₋₃₃₉₎ and CRT₍₁₋₃₅₁₎ are shown in Figure 2A [↗](#). SDS-PAGE gels corresponding to the purified proteins are shown in Figure 2B [↗](#). CRT_{Ins5} was not purified in sufficient quantities needed for ITC experiments and was not further investigated in this study.

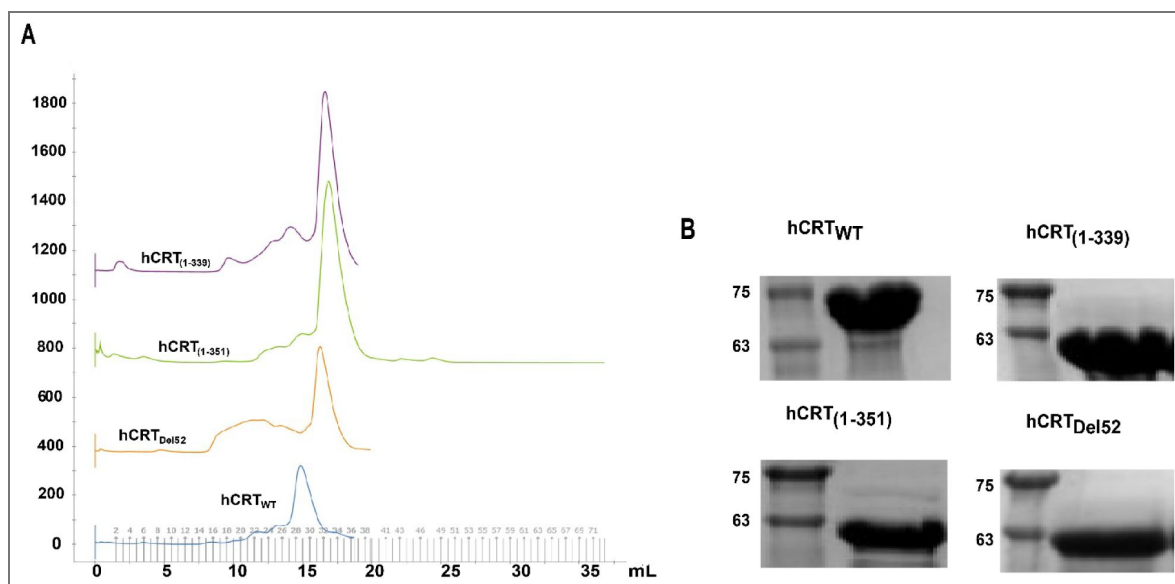


Figure 2. Purification of recombinant human wild-type CRT, CRT_{Del52}, and C-domain truncation mutants.

A) Representative chromatograms of recombinant GB1-and his-tagged human CRT_{WT} (hCRT_{WT}), CRT_{Del52}, and the C-domain truncation mutants hCRT₍₁₋₃₅₁₎ and hCRT₍₁₋₃₃₉₎ purified by gel exclusion chromatography after expression in *E. coli* and initial purification using a nickel resin. **B)** Representative Coomassie blue-stained SDS-PAGE gels of purified proteins that were used for calcium-binding measurements.

For the ITC measurements, calcium titrations were performed after pre-blocking the high affinity calcium binding sites with 50-100 μM CaCl_2 . Similar binding profiles and properties were observed with wild-type CRT, CRT_{Del52}, and the CRT₍₁₋₃₅₁₎ mutant (Figures 3A, C and D and Table 1). However, consistent with previous studies with mCRT₍₁₋₃₃₉₎, endothermic calcium binding by human CRT₍₁₋₃₃₉₎ was of lower magnitude compared to other constructs, with a K_D value that was too low to be reliably fit, suggesting the loss of low affinity calcium binding sites in this mutant (Figure 3B and Table 1). There are exothermic peaks observed with human CRT₍₁₋₃₃₉₎ at the beginning of the calcium titration, which may be attributed to altered low affinity sites retained within the mutant or reduced affinity of the high affinity sites. We concluded that CRT_{Del52} retains low affinity calcium binding that is measurable by ITC.

Similar ER calcium levels in CRT-KO HEK cells reconstituted with wild-type CRT or CRT_{Del52}

While the CRT_{Del52} mutant retains some ability to bind calcium (Figure 3 and Table 1), the loss of the ER-retention KDEL sequence induced by the frameshift mutation, which results in the secretion of the protein from the ER (Araki et al., 2016, Chachoua et al., 2016, Han et al., 2016, Arshad and Cresswell, 2018, Liu et al., 2020, Masubuchi et al., 2020, Venkatesan et al., 2021a, Pecquet et al., 2023) might be expected to cause ER calcium insufficiency in cells expressing CRT_{Del52} compared to wild-type CRT. We investigated ER calcium signals in CRT knock-out HEK293T cells generated by CRISPR/Cas9 gene editing. A clonal KO line was generated and subsequently reconstituted with constructs encoding wild-type CRT, CRT_{Del52}, or CRT_{Del52} with a KDEL sequence added at the C-terminus (CRT_{Del52}-KDEL). Protein expression was assessed by immunoblotting with anti-CRT(N), which detects an N-domain epitope present in all CRT constructs, and anti-CRT(C-mut), which was specifically raised against the mutant CRT C-terminus (Venkatesan et al., 2021a). While the recognition with anti-CRT (N) was very weak in lysates of cells expressing CRT_{Del52}, which is secreted, CRT_{Del52}-KDEL is readily detectable with both antibodies. Both wild-type CRT and CRT_{Del52}-KDEL were reconstituted to higher levels than in the parental cells (Figure 4A, lanes 1 and 2 compared to 7-8 and 11-12). The KO and reconstituted cells were further transfected with pCIS-GEM-CEPIA1er, which encodes a ratiometric ER calcium probe (Suzuki et al., 2014). Emission signals for calcium-bound (452-455 nm) and unbound (498 nm) probe were measured using flow cytometry in the Pacific Blue and AmCyan channels, respectively, following excitation with the 405 nm (violet) laser. Flow cytometric analysis was undertaken in a buffer containing 2 mM extracellular Ca^{2+} (Figures 4B and 4C). There was a significant decrease in basal free ER calcium, measured by the bound/unbound ratios in CRT-KO cells compared to the parental cells, (Figure 4D, left panel). ER calcium levels were partially restored, although non-significantly, when the CRT-KO cells were reconstituted with wild-type CRT. CRT-KO cells reconstituted with CRT_{Del52} and CRT_{Del52}-KDEL showed small but non-significant decreases in ER calcium levels compared to those reconstituted with wild-type CRT (Figure 4D, right panel). Changes to ER calcium levels in response to thapsigargin (Tg), a SERCA inhibitor, were similar in the parental vs. CRT-KO cells, and in the CRT-KO cells compared with those reconstituted with different CRT proteins (Figure 4E).

Similar cytosolic calcium and SOCE signals in a CRT-KO megakaryocyte cell line reconstituted with wild-type CRT or CRT_{Del52}

MPN CRT mutations cause thrombopoietin-independent differentiation of megakaryocytes to platelets via the thrombopoietin receptor (MPL) (Araki et al., 2016, Chachoua et al., 2016, Marty et al., 2016, Li et al., 2018, Masubuchi et al., 2020). We, therefore, used a megakaryoblast cell line, MEG-01, to assess the role of CRT deficiency and the CRT_{Del52} mutant on basal cytosolic calcium levels and SERCA inhibition-induced calcium influx into the cytosol. As with the HEK cells (Figure 4), we created a stable CRT knockout clone of MEG-01 cells by CRISPR/Cas9 editing for cytosolic calcium measurements. The cells were subsequently

Figure 3. Calcium binding to the low affinity sites of purified human CRT proteins.

Low affinity calcium binding by purified hCRT_{WT}, CRT_{Del52}, and the indicated truncation mutants was measured using isothermal titration calorimetry. **A-D)** Representative isotherm and fitting curves of calcium binding to the low affinity sites of the wild-type and mutant CRT proteins as indicated, measured using isothermal titration calorimetry (ITC). Prior to titration, the proteins were incubated with 50-100 μ M CaCl₂ to block the high affinity calcium binding sites. At 37°C, 10 mM CaCl₂ was titrated against 100 μ M protein. The heat change was measured for 25 injections, and the data were curve-fitted to calculate the dissociation constant, enthalpy, and binding stoichiometry using NanoAnalyzer software. Data replicates and statistical analyses are specified in Table 1.

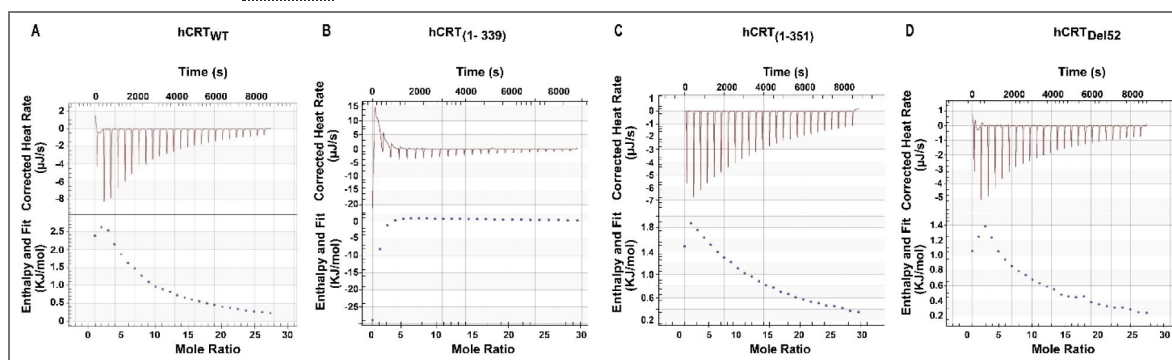


Table 1. Calorimetric data on measurement of calcium binding to the low affinity binding site of recombinant purified CRT.

Recombinant CRT proteins/Isothermal calorimetric measurement	ΔH (kJ/mol)	n (mol)	Dissociation constant - K_D (μ M)
CRT _{WT}	7.88 ± 2.15	5.40 ± 1.31	574.24 ± 109.93
CRT ₍₁₋₃₅₁₎	79.82 ± 32.55	2.59 ± 1.56	444.34 ± 139.99
CRT ₍₁₋₃₃₉₎			Too low to reliably fit
CRT _{Del52}	3.89 ± 1.36	7.73 ± 1.81	938.82 ± 275.89

Calorimetric data are represented as the mean $\pm$ S.E. Data shown are averages from 3 or 5 independent replicates.

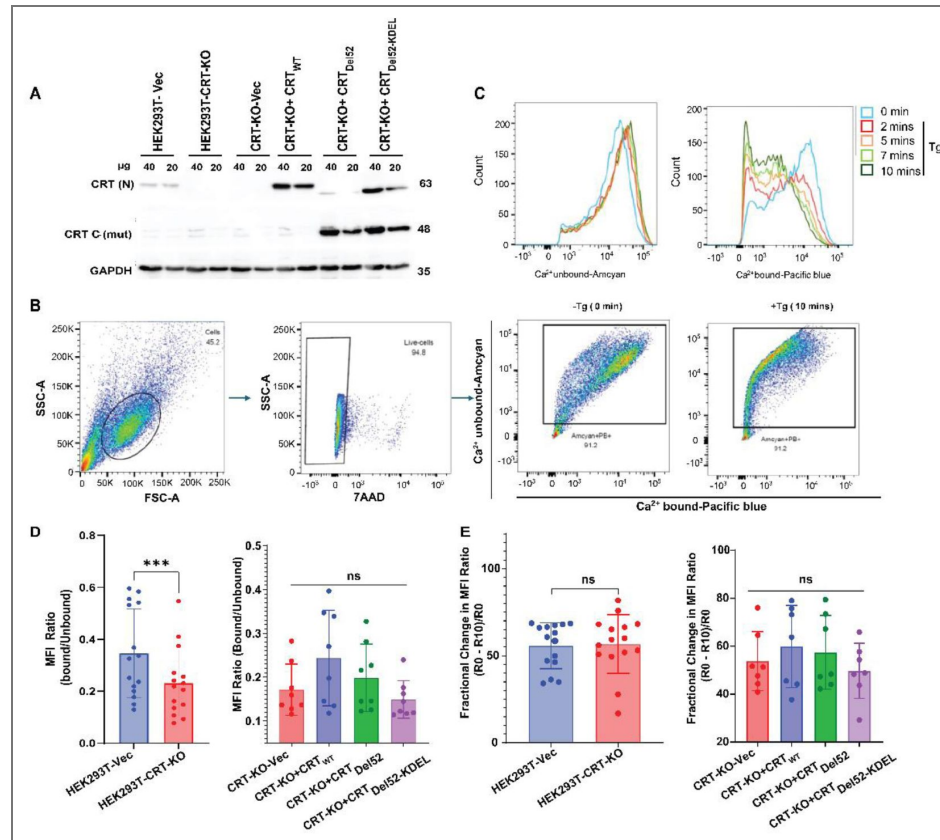


Figure 4. ER calcium measurements in CRT-KO HEK293T cells and those reconstituted with wild-type CRT, CRT_{Del52}, or CRT_{Del52-KDEL}.

A) CRISPR-Cas9 editing of HEK cells using lentiviral constructs of calreticulin-targeting gRNA (CRT-KO) or empty vector (HEK-Vec). CRT-KO HEK cells were subsequently reconstituted with wild-type human CRT, CRT_{Del52}, or CRT_{Del52} with a C-terminal KDEL sequence as indicated. Representative blots showing expression levels of CRT in the indicated HEK cells. Cell lysates were subjected to SDS-PAGE followed by immunoblotting. Wild-type CRT, mutant CRT, and GAPDH were detected with anti-CRT (N), anti-CRT C-(mut), and anti-GAPDH antibodies, respectively. **B)** Gating strategy for flow cytometry. HEK-Vec cells transiently transfected with the vector encoding the ratiometric GEM-CEPIA1er probe for 72 hours were used to assess ER calcium levels by flow cytometry based on the measurements of fluorescence emission signals for the Ca²⁺-bound or unbound probe at 452-455 nm and 498 nm, respectively following excitation with the 405 nm (violet) laser. Transfected cells were gated based on forward (FSC-A) and side scatter (SSC-A), then on 7AAD-negative (live) cells, and further on the cells positive for the calcium-bound or unbound probe signals. Representative dot plots of changes in fluorescence intensities in response to treatment with thapsigargin (Tg), a SERCA inhibitor, before (0 mins) or after (10 mins) treatment, are shown. **C)** Representative histograms showing changes to Ca²⁺-bound and unbound signals at different time points following Tg treatment of cells. Mean fluorescence intensities (MFI) for Ca²⁺-bound and unbound signals were used to calculate the bound/unbound ratios at different time points. **D and E)** The basal free ER calcium signals were measured as bound/unbound MFIs ratios from the corresponding histograms of CRT-KO or HEK-Vec cells (D, left panel) or CRT-KO cells reconstituted with the indicated CRT constructs (D, right panel). The fractional changes in ER calcium signals in response to Tg were measured at different time points in the indicated cells (E). The fractional change ratios were normalized to the baseline and calculated by subtracting the ratio at 10 min from the ratio at 0 mins (R0-R10/R0)×100. Statistical significance was determined using GraphPad Prism and based on paired t-tests (left panels of D and E) or one-way ANOVA analyses with Tukey's test (right panels of D and E). ns, not significant; *** P value < 0.001. All measurements were undertaken in the presence of 2 mM extracellular Ca²⁺. Data shown in D and E are based on 7-15 independent experiments as indicated by the individual data points within the graphs. GAPDH, glyceraldehyde-3-phosphate dehydrogenase; SDS-PAGE, sodium dodecyl sulfate-polyacrylamide gel electrophoresis.

reconstituted with wild-type CRT or CRT_{Del52}. Representative immunoblots are shown to demonstrate the successful knockout and reconstitution (Figure 5A). Wild-type CRT reconstitution in MEG-01 CRT-KO cells resulted in expression levels higher than those in the parental cells (Figure 5A, lanes 7 and 8 compared with lanes 1 and 2). In the presence of 2 mM Ca²⁺ in the media, we observed no significant differences in the basal cytosolic calcium levels or CPA-induced calcium release from ER in CRT-KO cells compared to the parental MEG-01 cells (Figure 5C and 5D). Additionally, no significant differences in basal cytosolic calcium were observed in CRT-KO cells compared to those reconstituted with wild-type CRT or CRT_{Del52} (Figure 5E). Furthermore, cytosolic calcium levels in CRT-KO and the two reconstituted cell lines were not statistically different following CPA treatments.

SOCE is an important process for restoring ER calcium and maintaining intracellular calcium levels. In contrast to cytosolic calcium signals in the presence of 2 mM Ca²⁺ in the media, in calcium-free saline, CRT-KO cells exhibited a small but significant increase in basal cytosolic calcium levels relative to the parental cells (Figure 6A, left panel and 6B). Upon ER calcium depletion with CPA, the mean fractional increase in the cytosolic calcium measured in the Ca²⁺-free saline did not differ significantly in parental cells vs. CRT-KO cells. The areas under the curve were also not different for the two cell lines (Figures 6A and 6C). However, consistent with the higher basal cytosolic calcium measured in CRT-KO cells in calcium-free saline, the measured SOCE signal (following the addition of 2 mM Ca²⁺ and CPA) was significantly reduced in CRT-KO cells compared to parental cells (Figure 6D). Reconstitution of the CRT-KO cells with either wild-type CRT or CRT_{Del52} tended to reduce basal cytosolic calcium and induce SOCE (Figure 6A, right panel, 6E and 6G), but both trends were non-significant compared to CRT-KO cells. Importantly, there were no significant differences between the wild-type CRT and CRT_{Del52} reconstituted cells, in either their basal or CPA-induced cytosolic calcium signals or in SOCE (Figures 6E-G).

CRT deficiency induced numerous transcriptional changes to genes within the calcium signaling pathway

As noted in Figures 4 and 6, some of the measured differences between CRT-KO and parental cells are not fully reversed by reconstitution with wild-type CRT, although the trends are maintained. We predicted that there may be other indirect effects of CRT-KO and reconstitution that affect cellular calcium levels and calcium signaling that could explain some of these measured differences. As proof of concept, we undertook bulk RNA-sequencing to determine differential gene expression in MEG-01 parental vs. CRT-KO cells. There were 3078 upregulated and 1790 downregulated genes. The volcano plot and heat map show identified genes that were differentially expressed in CRT-KO MEG-01 compared to parental cells (Figure 7A). Based on pathway analysis, numerous genes involved in cellular calcium signaling were identified. Of specific note is the upregulation of GRP94 encoded by the *HSP90B1* gene, BiP encoded by *HSPA5* gene, and PDIA6 encoded by *PDIA6* gene, which are known to have low affinity calcium binding sites (Lievremont et al., 1997, Macer and Koch, 1988, Biswas et al., 2007, Marzec et al., 2012, Okumura et al., 2021) (Figures 7A and 7B). Other studies have reported similar upregulation of these genes in CRT-KO (Tang et al., 2025) as well as in heterozygous CRT_{Del52} knock-in cells (Figure 7C) (Fosselteder et al., 2023). Upregulation of these proteins may serve a compensatory role in buffering calcium in CRT-KO cells. Additionally, PDIA3, which is shown to regulate the activities of stromal interaction molecule 1 (STIM1) (Prins et al., 2011) and Sarco/endoplasmic reticulum Ca²⁺ ATPase (SERCA2b) (Li and Camacho, 2004), was consistently upregulated in all the studies.

We further used real-time PCR with target gene primers (Supplementary Table 2) to validate the expression of the key ER resident proteins identified from the RNA-sequencing data which have roles in calcium binding and regulation of ER calcium homeostasis. Consistent with the RNA-seq data, the expression of *HSPA5*, *HSP90B1*, *PDIA6* and *PDIA3* mRNA transcripts was significantly increased in CRT-KO MEG-01 compared to parental cells. Reconstitution of CRT-KO cells with wild-type CRT but not CRT_{Del52} showed a significant decrease in the expression of *HSPA5*, *HSP90B1*, *PDIA6* and *PDIA3* (Figures 7D-G). Protein levels of BiP, GRP94, PDIA6 and PDIA3 were also

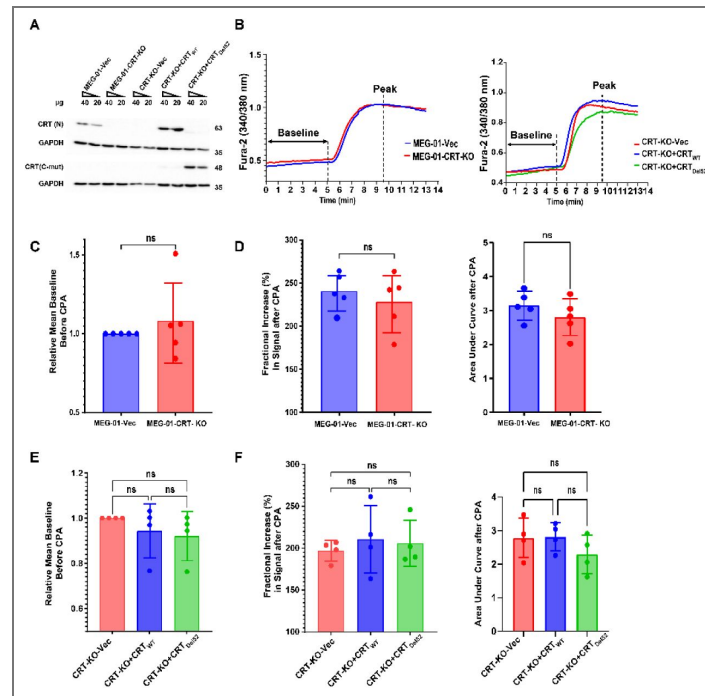


Figure 5. Measurements of cytosolic calcium levels in megakaryoblastic (MEG-01) parental cells, CRT-KO cells, or those reconstituted with wild-type CRT or CRT_{Del52}

Cells in 2 mM extracellular Ca^{2+} were loaded with the Fura-2/AM dye, and the ratio of the signals was calculated following excitation at 340 and 380 nm and emission at 510 nm. **A)** CRISPR-Cas9 editing of MEG-01 cells using lentiviral constructs encoding calreticulin-targeting gRNA (CRT-KO) or empty vector (Vec). CRT-KO MEG-01 cells were subsequently reconstituted with viral constructs encoding wild-type human CRT or CRT_{Del52} or transduced with empty vector (Vec) as indicated in the Figure panels. Representative immunoblots of CRT expression in the indicated cells detected by the anti-CRT(N) and anti-CRT(C-mut) antibodies. **B)** Representative calcium imaging data from indicated cells in the presence of 2 mM extracellular Ca^{2+} and following ER calcium depletion with SERCA inhibitor, cyclopiazonic acid (CPA). **C and E)** Basal cytosolic calcium levels were plotted as the means of ratios of emission signals following excitation at 340 and 380 nm. **D and F)** Fractional changes to cytosolic calcium levels in the indicated cell lines, measured following treatment with CPA and total calcium release determined as the area under the curve (AUC), are plotted relative to the baseline following CPA addition. Data shown in C-F are based on 4 or 5 independent measurements, each including 50-100 cells. A paired t-test was used to analyze the mean difference between two groups (C and D). Multiple comparisons among groups were analyzed using one-way ANOVA analyses with Tukey's test (E and F). Statistical significance was determined based p-value ≤ 0.05 using GraphPad Prism. ns, not significant.

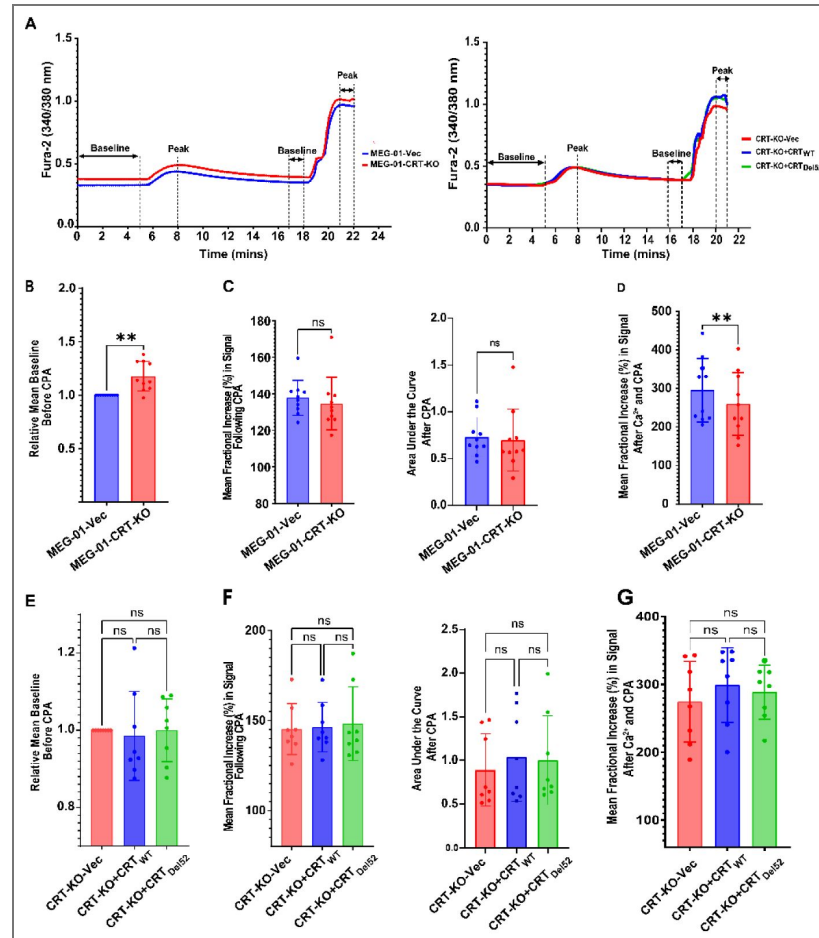


Figure 6. Measurements of cytosolic calcium levels and SOCE in MEG-01 parental cells, CRT-KO, or those reconstituted with wild-type CRT or CRT_{Del152}.

Cells were loaded with Fura-2 dye, and the ratio of the 340 nm/380 nm signals was calculated following excitation at 340 and 380 nm and emission at 510 nm. **A)** Representative calcium imaging data from cells in the absence of extracellular calcium, followed by the addition of CPA, and subsequent addition of extracellular calcium with CPA as indicated. **B and E)** Compiled basal cytosolic calcium levels in the indicated cells in the absence of extracellular calcium plotted as the ratios of 340 nm/380 nm signals. **C and F)** Fractional changes to cytosolic calcium and total calcium release measured as the area under the curve (AUC) in the indicated cells following ER calcium depletion with the CPA. **D and G)** Fractional changes to cytosolic calcium following ER calcium depletion with the CPA and subsequent addition of extracellular calcium with CPA. Data are plotted as a fractional increase relative to baseline following CPA addition. SOCE in the indicated cells is plotted as the fractional increase in signal following the addition of 2 mM extracellular calcium in the presence of CPA. Data shown are based on 8-10 independent measurements, each with 20-100 cells. Comparisons were performed by paired t-test and one-way ANOVA analysis using GraphPad Prism. Statistical significance is based on a p-value ≤ 0.05 . ns, not significant.

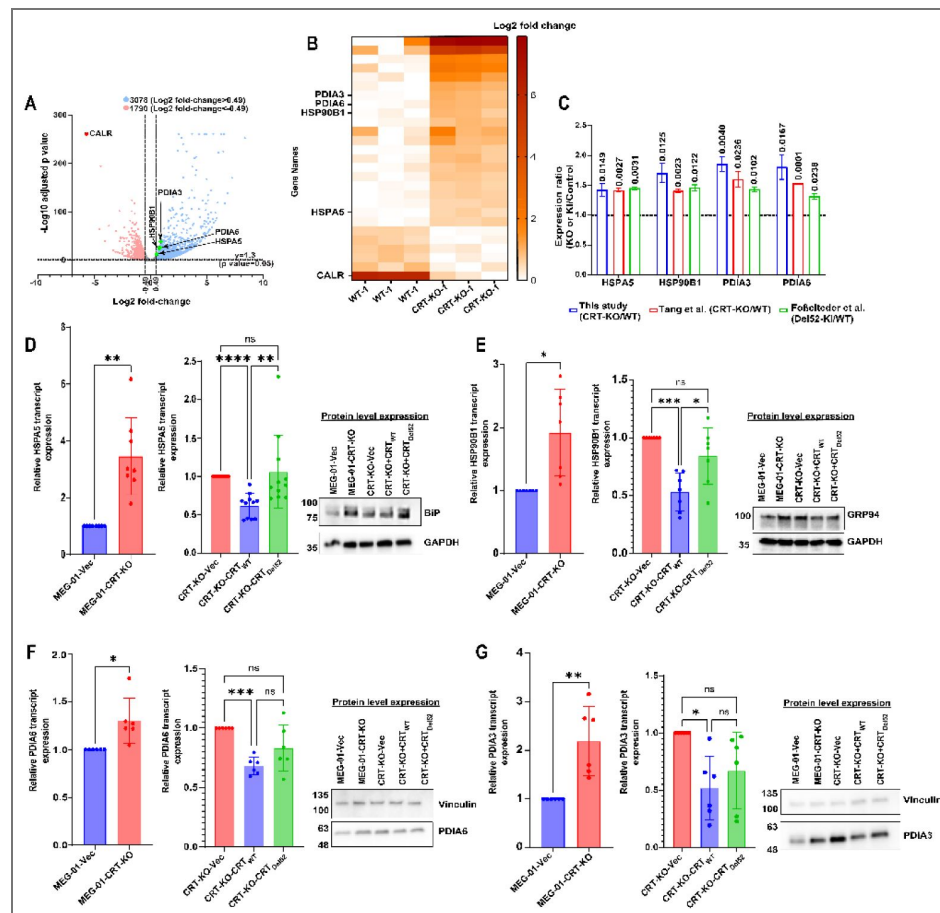


Figure 7. Transcriptional changes to the expression of genes encoding ER calcium binding/regulating proteins in CRT-KO MEG-01 cells.

A) RNA sequencing analysis was used to identify differentially expressed genes in Meg-01 CRT-KO cells in comparison to parental Meg-01 cells. Volcano plot shows significantly upregulated (blue) and downregulated genes (red) genes identified by RNA sequencing. Genes related to ER calcium binding and homeostasis have been highlighted and labeled. The differentially expressed genes were selected as those with a log2 fold-change of <-0.49 or >0.49 and p values <0.05. **B)** Heat map shows the differential expression of genes encoding ER resident calcium binding proteins or ER calcium regulators that were identified by RNA sequencing in CRT-KO Meg-01 cells compared to parental cells (n=3 biological replicates). **C)** Comparison of expression ratio of *HSPA5*, *HSP90B1*, *PDIA3* and *PDIA6* in CRT-KO Meg-01 cells (this study) or murine lung cancer cells (Tang et al., 2025) or heterozygous CRT^{Del52} knock-in (Del52-KI) cells (Fosselteder et al., 2023) normalized to their expression in respective control cell lines expressing wild type CRT. The normalized expression of various genes identified in each dataset was calculated as a ratio of expression in CRT-KO or CRT^{Del52} knock-in cells relative to the expression in wild-type CRT-expressing cells. Multiple unpaired t-tests were used to determine statistical significance with the Benjamini-Hochberg FDR correction (p-adjusted) <0.05. **D-G)** Left and middle panels: Relative expression of mRNA transcripts of *HSPA5* (n=8-11), *HSP90B1* (n=7), *PDIA6* (n=6) and *PDIA3* (n=6) genes in the indicated MEG-01 cells determined by RT-qPCR using specific gene primers. Relative gene expression was calculated by normalizing the target gene's cycle threshold (Ct) to a single or multiple endogenous controls (ACTB, GAPDH and HPRT1) which was used for comparison across different cell lines as indicated. Data show 6 - 11 independent runs with duplicate measurements for each run. Student t-test was used to determine the statistical significance between two groups while multiple group comparison was performed by one-way ANOVA analysis using GraphPad Prism. Statistical significance is based on a p-value ≤ 0.05. ns, not significant. Right panels: Representative immunoblots of BiP, GRP94, PDIA6 and PDIA3 expression in the indicated cells detected by specific antibodies.

assessed in the indicated cells by immunoblots using specific antibodies (supplementary methods). Representative immunoblots show increased expression of BiP (Figure 7D, right panel [↗](#)), GRP94 (Figure 7E, right panel [↗](#)), PDIA6 (Figure 7F, right panel [↗](#)) and PDIA3 (Figure 7G, right panel [↗](#)) in CRT-KO cells compared to parental MEG-01 cells as well as increased expression in CRT-KO-Vec cells compared to CRT-KO cells reconstituted with wild type CRT. The effects of CRT_{Del52} in reducing expression were generally less robust than wild type CRT.

Discussion

Calreticulin is a ubiquitously expressed ER luminal calcium-binding chaperone, with a role in ER calcium storage (Venkatesan et al., 2021b [↗](#), Michalak, 2024 [↗](#)). Previous studies using dye-based assays reported impaired calcium binding by the MPN-associated CRT_{Del52} mutant (Ibarra et al., 2022 [↗](#)). Yet direct and quantitative analyses of its calcium-binding properties have been lacking. In this study, we investigated low affinity calcium binding by wild-type human CRT, CRT_{Del52}, and the truncation mutants CRT₍₁₋₃₅₁₎ and CRT₍₁₋₃₃₉₎ using ITC (Figure 3 [↗](#) and Table 1 [↗](#)). Our studies show that wild-type CRT, CRT_{Del52}, and CRT₍₁₋₃₅₁₎ retain low affinity calcium binding sites, exhibiting average K_D values of 574.24, 938.82, and 444.34 μ M, respectively. On the other hand, low affinity calcium binding by CRT₍₁₋₃₃₉₎ could not be reliably fit. This is the first quantitative characterization of the low affinity calcium binding profiles of wild-type human CRT, CRT_{Del52}, and functionally informative truncation mutants.

Although CRT_{Del52} retains at least partial low affinity calcium binding capability, loss of its KDEL retention sequence (Figure 1 [↗](#)) could still disrupt ER calcium storage and signaling. Interestingly, the cellular assays reveal that ER and cytosolic calcium levels, as well as SOCE, are similar between CRT-KO cells and those reconstituted with either wild-type CRT or CRT_{Del52} (Figures 4-6 [↗](#)). One study has reported decreased ER calcium levels in a human osteosarcoma cell line expressing CRT_{Del52} compared to those expressing wild-type CRT, following transient transfections (Ibarra et al., 2022 [↗](#)). The same study suggested the contributions of the P and C-domain of wild-type in restoring ER calcium levels in cells expressing CRT_{Del52} (Ibarra et al., 2022 [↗](#)). While the conditions of those experiments differ from the studies described here which use a CRT-KO background, our findings are consistent with the prediction that any excess free ER calcium generated in cells lacking CRT or sufficient ER calcium binding sites would rapidly diffuse out of the ER and be pumped out of the cell.

Related to cytosolic calcium levels and SOCE activation, consistent with our findings (Figure 6G [↗](#)), a previous study found no differences in cytosolic calcium levels or SOCE induction in myeloid progenitor 32D cells expressing wild-type CRT and CRT_{Del52} both under conditions of cytokine deprivation and thrombopoietin (TPO) stimulation (Bhuria et al., 2024 [↗](#)). While cultured megakaryocytes from MPN patients with *CALR* mutations have SOCE abnormalities compared to healthy control cells (Pietra et al., 2016 [↗](#), Di Buduo et al., 2020 [↗](#)) sample to sample heterogeneities in primary cells make it difficult to attribute measured differences solely to calcium binding by CRT or its mutants.

We found that CRT-KO HEK cells displayed lower ER calcium compared with the parental HEK cells (Figure 4D [↗](#), left panel) and also that CRT-KO Meg-01 cells displayed increased cytosolic calcium in the absence of extracellular calcium (Figure 6B [↗](#)). However, reconstitutions with wild-type CRT did not reverse these effects (Figure 4D [↗](#), right panel and Figure 6E [↗](#)). Transcriptomic analysis identified significant alterations in the expression of genes related to calcium storage and signaling between parental and CRT-KO Meg-01 cells (Figure 7 [↗](#)). These changes highlight adaptive or compensatory pathways induced by CRT-KO and emphasize that, when comparing wild-type and CRT-KO cell lines for subcellular calcium levels or responses to SERCA inhibitors, the measured effects might not be based solely on the abilities of CRT to maintain ER and cytosolic Ca^{2+} levels via its calcium binding, storage and buffering capacity. Rather, many other compensatory cellular changes could affect subcellular calcium levels and signaling. In this study, various ER proteins with known calcium binding activities were found to be upregulated in CRT-KO cells (Figure 7A [↗](#) and 7B [↗](#)). GRP94 (Macer and Koch, 1988 [↗](#), Biswas et al., 2007 [↗](#), Marzec et al., 2012 [↗](#)), BiP (Lievremont et al., 1997 [↗](#)) and our unpublished findings, and PDIA6 (Okumura

et al., 2021 [↗](#)) are each known to contain low affinity calcium binding sites. CRT-KO cells have significantly induced expression of the genes encoding these proteins. Wild type CRT reconstitution into the CRT-KO cells significantly reduced the expression of these ER proteins, whereas CRT_{Del52} reconstitution only partly and non-significantly reduced expression of *HSP90B1* and *PDIA6* and did not reduce the expression of *HSPA5* (Figures 7D-F [↗](#)). GRP94, BiP and PDIA6 have functions in ER protein folding and quality control (Hendershot et al., 2024 [↗](#), Okumura et al., 2021 [↗](#)). It is known that CRT_{Del52} has impaired chaperone activity towards its substrates compared with wild type CRT (Arshad and Cresswell, 2018 [↗](#), Schurch et al., 2022 [↗](#), Kaur et al., 2025 [↗](#)). This could account for some of these measured differences between wild type CRT and CRT_{Del52} in altering the expression of the ER chaperones and folding factors. However, their induction in knock-in cells with heterozygous expression of wild type CRT and CRT_{Del52} (Fosselteder et al., 2023 [↗](#)), in which the expressed wild type CRT appears sufficient for multiple protein folding activities (Kaur et al., 2025 [↗](#), Schurch et al., 2022 [↗](#)) could point to compensatory effects relevant to calcium signaling. Regardless of the precise mechanisms relevant to induction of the calcium binding proteins, their combined expression, despite protein compositional differences between the KO and reconstituted cells, appears sufficient to maintain ER calcium levels (Figure 4D [↗](#), right panel), calcium release from the ER (Figure 4E [↗](#), right panel, and 5F) and cytosolic calcium levels (Figure 5E [↗](#)).

PDIA3 (ERp57) forms a complex with CRT and interacts with and regulates the activities of the ubiquitously expressed SERCA2b (Li and Camacho, 2004 [↗](#)). Its expression is also induced by the CRT-KO and suppressed by wild type CRT expression but non-significantly by CRT_{Del52} expression. Altered SERCA2b activity could also affect basal ER calcium levels, contributing to similarities between CRT-KO, and wild type CRT or CRT_{Del52} reconstituted cells.

Collectively, our findings reveal that CRT_{Del52} retains some ability to bind calcium at low affinity sites. CRT-KO cells reconstituted with wild-type or CRT_{Del52} have comparable ER and cytosolic calcium levels and store-operated calcium entry (SOCE). CRT-KO induces broad changes in ER calcium binding protein expression as well as in the expression of ERp57. Overall, we can conclude that a number of factors that contribute to the maintenance of ER calcium levels and regulation of calcium transport into the ER are altered in CRT-KO cells. Thus, any disease-related changes in calcium signaling and homeostasis are likely multifactorial, encompassing transcriptional regulation and altered expression of critical mediators of calcium storage, transport and signaling pathways. These studies provide a more nuanced understanding of CRT's role in cellular calcium signaling and may inform future therapeutic strategies targeting calcium signaling pathways in hematological malignancies.

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Additional files

[Supplemental Methods](#) [↗](#)

Additional information

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Peer reviews

Reviewer #1 (Public review):

The authors attempted to compare calcium calcium-binding properties of wildtype calreticulin with calreticulin deletion mutant (CRTDel52) associated with myeloproliferative neoplasms.

The researchers conducted their study using advanced techniques. They found almost no difference in calcium binding between the two proteins and observed no impact on calcium signaling, specifically store-operated calcium entry (SOCE). The study also noted an increase in ER luminal calcium-binding chaperone proteins. Surprisingly, the authors selected flow cytometry as a technique for measurements of ER luminal calcium. Considering the limitations of this approach, it would be better to use alternative approaches. This is particularly important as previous reports, using cells from MPN patients, indicate reduced ER luminal calcium and effects on SOCE (Blood, 2020). This issue matters because earlier research with MPN patient cells reported reduced ER luminal calcium levels and altered SOCE (Blood, 2020). How do the authors explain the difference between their results and previous findings about lower ER luminal calcium and changed SOCE in MPN patient cells expressing CRTDel52? Other studies have found that unfolded protein responses are activated in MPN cells with CRTDel52 calreticulin (see Blood, 2021), and increased UPR could account for higher levels of some ER-resident calcium-binding proteins observed here. Overall, it remains unclear how this work improves our understanding of MPN or clarifies calreticulin's role in MPN pathophysiology.

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Reviewer #2 (Public review):

Summary:

Tagoe and colleagues present a thorough analysis of the calcium (Ca^{2+}) binding capacity of calreticulin (CRT), an endoplasmic reticulum (ER) Ca^{2+} -buffer protein, using a mutant version (CRT del52) found in myeloproliferative neoplasms (MPNs). The authors use purified human CRT protein variants, CRT-KO cell lines, and an MPN cell line to elucidate the differing Ca^{2+} dynamics, both on the level of the protein and on cell-wide Ca^{2+} -governed processes. In sum, the authors provide new insights into CRT that can be applied to both normal and malignant cell biology.

First, the authors purify CRT protein and perform isothermal titration calorimetry to quantify the Ca^{2+} binding capacity of CRT. They use full-length human CRT, CRT del52, and two truncations of CRT (1-339 and 1-351, the former of which should lead to the entire loss of low-affinity Ca^{2+} binding). While CRT del52 has previously been shown to lead to a decrease in Ca^{2+} binding affinity in other models, the ITC data show that this is retained in CRT del52.

Next, the authors utilize a CRT-KO cell line with subsequent addition of CRT protein variants to validate these findings with flow cytometric analysis. Cells were transfected with a ratiometric ER Ca^{2+} probe, and fluorescence indicates that CRT del52 is unable to restore basal ER Ca^{2+} levels to the same extent as CRT wild-type. To translate these findings to MPNs, the authors perform CRT-KO in a megakaryocytic cell line, where reconstitution with either CRT variant did not cause a difference in cytosolic calcium levels. The authors further test store-operated calcium entry (SOCE), an important process for maintaining ER Ca^{2+} levels, in these cells, and find that CRT-KO cells have lower SOCE activity, and that this can be slightly recovered with CRT addition.

Finally, the authors ask whether other effects of CRT-KO/reconstitution can affect the cellular Ca^{2+} signaling pathway and levels. RNASeq analysis revealed that CRT-KO leads to an increase in various chaperone protein expressions, and that reconstitution with CRT del52 is unable to reduce expression to the same extent as reconstitution with CRT wildtype.

Strengths:

The authors provide new insights into CRT that can be applied to both normal and malignant cell biology.

Weaknesses:

(1) The authors should consider discussing the high-affinity Ca^{2+} binding site more in the introduction. Can they show a proof-of-concept experiment that validates that incubation of recombinant CRT reduces the function of that high-affinity Ca^{2+} binding site?

(2) For Figure 2B, do you have an explanation for why the purified proteins run higher than predicted (48-52kDa) - are these proteins still tagged with pGB1?

(3) The MEG-01 cell line has the BCR::ABL1 translocation, while CRT mutations are strictly found in BCR::ABL1 negative MPNs. Could these experiments be repeated in these cells treated with imatinib to decrease these effects, or see if basal MEG-01 Ca^{2+} levels/activity are changed with or without imatinib?

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Author response:

Public Reviews:

Reviewer #1 (Public review):

The researchers conducted their study using advanced techniques. They found almost no difference in calcium binding between the two proteins and observed no impact on calcium signaling, specifically store-operated calcium entry (SOCE). The study also noted an increase in ER luminal calcium-binding chaperone proteins. Surprisingly, the authors selected flow cytometry as a technique for measurements of ER luminal calcium. Considering the limitations of this approach, it would be better to use alternative approaches.

The flow cytometric assay shows good responsiveness to conditions expected to alter ER calcium levels (Figure 4C), is high throughput compared to microscopy, and allows for averaging of signals across a large number of cells. This was thus our original method of choice.

This is particularly important as previous reports, using cells from MPN patients, indicate reduced ER luminal calcium and effects on SOCE (Blood, 2020). This issue matters because earlier research with MPN patient cells reported reduced ER luminal calcium levels and altered SOCE (Blood, 2020). How do the authors explain the difference between their results and previous findings about lower ER luminal calcium and changed SOCE in MPN patient cells expressing CRTDel52?

We thank the reviewer for asking for these clarifications. The referenced study (Di Buduo et al. Blood, 135(2):133-143, 2020) first showed that thrombopoietin induces spontaneous cytosolic calcium spikes in cultured megakaryocytes, which is dependent on store operated calcium entry (SOCE). In parallel, STIM1-ORAI interactions were induced by thrombopoietin. On the other hand, the addition of thrombopoietin caused the dissociation of STIM1-calreticulin interactions, based on proximity ligation assays. The implication is that signaling via the thrombopoietin receptor (TPOR/MPL) activation induces the dissociation of calreticulin-STIM1 complexes, and the formation of STIM1-ORAI complexes, which contribute to the measured spontaneous cytosolic calcium spikes. Different MPN mutations induced spontaneous calcium spikes in a thrombopoietin-independent manner, including the JAK2V617F mutations and the CALR type I and type II mutations. The study found that the number of megakaryocytes exhibiting spontaneous calcium spikes was enhanced in the context of both type I and type II CALR mutations compared to the JAK2V617F mutant. Correspondingly, the calreticulin-STIM1 interactions/cell were more significantly reduced for type I and type II CALR mutations compared to the JAK2V617F mutant. It was suggested that defective interactions between mutant calreticulin, ERp57, and STIM1 activated SOCE and generated spontaneous cytosolic calcium spikes. However, based on the findings with thrombopoietin, the spontaneous calcium spikes could simply result from thrombopoietin-independent MPL activation by the mutant calreticulin and JAK2V617F and downstream signaling. Importantly, the referenced studies did not directly measure ER luminal calcium. A number of undefined factors could account for the measured differences between the megakaryocytes from patients with calreticulin mutations vs. JAK2V617F. These include the relative mutant allele burdens, the extent of MPL activation, as well as genetic differences unrelated to calreticulin. Different from these experiments, through the use of purified proteins, our studies show that the Del52 mutant has calcium binding characteristics resembling that of the wild type protein. Additionally, through genetic manipulations in cell lines, our studies directly address the effects of calreticulin KO and its Del52 mutation upon ER luminal and cytosolic calcium levels, and cellular SOCE signals. We did not measure significant differences in any of these parameters between the KO cells and those

reconstituted with wild type calreticulin or the Del52 mutant. As noted by the editors, these results show that Ca^{2+} binding by calreticulin and store-operated Ca^{2+} entry in a cell are not fundamentally impacted by the type I deletion mutation. On the other hand, in primary megakaryocytes, when co-expressed with MPL, the Del52 mutant, through its known ability to bind and activate TPOR/MPL, is expected to induce SOCE and calcium fluxes similar to those induced by thrombopoietin. These points will be clarified in the revised discussion.

Other studies have found that unfolded protein responses are activated in MPN cells with CRTDel52 calreticulin (see Blood, 2021), and increased UPR could account for higher levels of some ER-resident calcium-binding proteins observed here.

Multiple studies have suggested the induction of the unfolded protein response (UPR) in cells expressing MPN mutants of calreticulin. We don't know the specific signals that cause the upregulation of various calcium binding proteins in calreticulin-KO cells and cells expressing the Del52 mutant. Indeed, these could result from increased protein misfolding in cells with wild type calreticulin deficiency. Alternatively, the sensing of cellular calcium perturbations could induce their expression. Regardless of the precise mechanisms underlying the expression changes in calcium binding proteins, the upregulated factors are predicted to compensate for calreticulin deficiency and contribute to the maintenance of the overall cellular calcium homeostasis. These points will be clarified in the revised discussion.

Overall, it remains unclear how this work improves our understanding of MPN or clarifies calreticulin's role in MPN pathophysiology.

The points discussed above as well as their implications for the understanding of calreticulin's role in MPN pathophysiology will be clarified in the revised manuscript.

Reviewer #2 (Public review):

Tagoe and colleagues present a thorough analysis of the calcium (Ca^{2+}) binding capacity of calreticulin (CRT), an endoplasmic reticulum (ER) Ca^{2+} -buffer protein, using a mutant version (CRT del52) found in myeloproliferative neoplasms (MPNs). The authors use purified human CRT protein variants, CRT-KO cell lines, and an MPN cell line to elucidate the differing Ca^{2+} dynamics, both on the level of the protein and on cell-wide Ca^{2+} -governed processes. In sum, the authors provide new insights into CRT that can be applied to both normal and malignant cell biology.

First, the authors purify CRT protein and perform isothermal titration calorimetry to quantify the Ca^{2+} binding capacity of CRT. They use full-length human CRT, CRT del52, and two truncations of CRT (1-339 and 1-351, the former of which should lead to the entire loss of low-affinity Ca^{2+} binding). While CRT del52 has previously been shown to lead to a decrease in Ca^{2+} binding affinity in other models, the ITC data show that this is retained in CRT del52.

Next, the authors utilize a CRT-KO cell line with subsequent addition of CRT protein variants to validate these findings with flow cytometric analysis. Cells were transfected with a ratiometric ER Ca^{2+} probe, and fluorescence indicates that CRT del52 is unable to restore basal ER Ca^{2+} levels to the same extent as CRT wild-type. To translate these findings to MPNs, the authors perform CRT-KO in a megakaryocytic cell line, where reconstitution with either CRT variant did not cause a difference in cytosolic calcium levels. The authors further test store-operated calcium entry (SOCE), an important process for maintaining ER Ca^{2+} levels, in these cells, and find that CRT-KO cells have lower SOCE activity, and that this can be slightly recovered with CRT addition.

Finally, the authors ask whether other effects of CRT-KO/reconstitution can affect the cellular Ca^{2+} signaling pathway and levels. RNASeq analysis revealed that CRT-KO leads to an increase in various chaperone protein expressions, and that reconstitution with

CRT del52 is unable to reduce expression to the same extent as reconstitution with CRT wildtype.

Strengths:

The authors provide new insights into CRT that can be applied to both normal and malignant cell biology.

We thank the reviewer for the recognition that this study is important for our understanding of both normal and malignant cell biology.

Weaknesses:

(1) The authors should consider discussing the high-affinity Ca²⁺ binding site more in the introduction. Can they show a proof-of-concept experiment that validates that incubation of recombinant CRT reduces the function of that high-affinity Ca²⁺ binding site?

In a previous study (Wijeyesakere et al. 2011 J. Biol Chem, 286 8771-8785), we showed that at a starting calcium concentration of 0 mM and with 3.3 mM injections of CaCl₂, the measured K_D value was 16.6 mM for calcium binding to wild type murine calreticulin, (which has ~95% % sequence identity with human calreticulin), corresponding to the high affinity site. On the other hand, at a starting calcium concentration of 50 mM and with 33 mM CaCl₂ injections, the measured K_D value for calcium binding to wild type murine calreticulin was 590 mM (corresponding to the low affinity sites). Thus, we did not measure the high affinity sites when the starting calcium concentration was 50 mM. This point will be clarified in the revised manuscript.

(2) For Figure 2B, do you have an explanation for why the purified proteins run higher than predicted (48-52kDa) - are these proteins still tagged with pGB1?

Yes, the purified proteins shown in Figure 2B retained a GB1 tag. This point will be clarified in the revised manuscript.

(3) The MEG-01 cell line has the BCR:ABL1 translocation, while CRT mutations are strictly found in BCR:ABL1 negative MPNs. Could these experiments be repeated in these cells treated with imatinib to decrease these effects, or see if basal MEG-01 Ca²⁺ levels/activity are changed with or without imatinib?

Thank you for the important point. We will assess cytosolic calcium levels in MEG-01 cells with or without imatinib.

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